JPetStore Modernization: Prompt Log by Milestone

Evidence for the Context Building, AI Guardrails and Tokenomics areas of the assessment.

Tool: Antigravity IDE (Google AI Pro plan) | Source: JPetStore 6 | Target: FastAPI, SQLAlchemy 2.0, Jinja2

1. How to read this log

Every prompt used to drive the AI-assisted build is listed in order, reworded for clarity and tagged to a milestone. Two working sessions are covered.

- **Session 1** (32 prompts): analysis, decisions, foundation and the first three phases. Compiled from the session's own summary.
- **Session 2** (34 prompts): Phase 4, hardening, documentation and handover. Extracted from the session transcript.

Type codes: **Dir** direction or decision, **Q** question, **Doc** documentation request, **Defect** a failure or gap I reported (each one is an iteration toward a green build), **Review** a request to audit or verify.

Milestone tags

Tag	Milestone
M0	Discovery and architecture decomposition
M1	Strategy, decisions and ADRs
M2	Foundation, configuration and dependency resolution
M3	Catalog and UI rendering
M4	Cart, preferences and visual parity
M5	Account, security and localization
M6	Orders and checkout
M7	AI-native layer (search, agent tools, pet advisor)

Tag	Milestone
M8	Verification, documentation and deployment
M9	Submission and handover

2. Prompt count by milestone

Tag	Session 1	Session 2	Total
M0 Discovery	3	0	3
M1 Strategy and ADRs	9	0	9
M2 Foundation	4	0	4
M3 Catalog and UI	1	0	1
M4 Cart and parity	3	0	3
M5 Account and security	4	4	8
M6 Orders	3	0	3
M7 AI-native layer	2	5	7
M8 Verification and docs	2	18	20
M9 Submission	1	7	8
Total	**32**	**34**	**66**

3. Session 1 prompts

ID	Prompt (reworded)	Tag	Type	Outcome
S1-01	Provided the assessment PDF as the project brief	M0	Dir	Context and evaluation criteria established
S1-02	Can the existing front end or its components be reused?	M1	Q	Recommended a 1:1 port of the JSPs to Jinja2, reusing CSS and 38 images

ID	Prompt (reworded)	Tag	Type	Outcome
S1-03	Explain the current product architecture	M0	Q	Controllers, services, MyBatis mappers and HSQLDB analyzed
S1-04	Prepare a decomposition document	M0	Doc	Architecture decomposition document
S1-05	Decision: keep the existing front end, update the backend, plan for AI-native and for vertical and horizontal scaling	M1	Dir	Stateless session and async runtime direction set
S1-06	Is a solution that scales both ways a good idea?	M1	Q	Trade-offs assessed, avoiding premature infrastructure
S1-07	Is any caching needed?	M1	Q	Read and write asymmetry assessed
S1-08	Clarification: data caching only, not resource caching	M1	Dir	Cache-aside for catalog data; inventory and order sequence never cached
S1-09	Update the roadmap with these decisions; can any AI capability be added?	M1	Doc	Roadmap updated; three AI features proposed
S1-10	Is SQLAlchemy the best option or are there others?	M1	Q	ORMs compared (SQLAlchemy, SQLModel, Tortoise)
S1-11	Choose mature, battle-tested frameworks that support the decisions	M1	Dir	FastAPI, Uvicorn, SQLAlchemy 2.0, Pydantic v2, Jinja2, Argon2id selected
S1-12	Create ADRs for all decisions made	M1	Doc	Architecture decision records (ADR-001 to ADR-008)
S1-13	Configuration approach (/config)	M2	Dir	Pydantic settings design for app configuration
S1-14	Build failure: C-extension compile error for pyyaml (missing MSVC build tools)	M2	Defect	Replaced binary-build dependencies with pure-Python alternatives
S1-15	Runtime failure: SQLAlchemy asyncio needs greenlet	M2	Defect	greenlet installed

ID	Prompt (reworded)	Tag	Type	Outcome
S1-16	Assess the full dependency tree and resolve it once	M2	Review	Requirements audited to remove native build needs
S1-17	Runtime failure: TypeError, unhashable dict as key in TemplateResponse	M3	Defect	Updated to the newer Starlette TemplateResponse signature
S1-18	Proceed to the next step	M6	Dir	Order checkout pipeline and account management started
S1-19	Cart screen layout is misaligned	M4	Defect	CSS float collapse fixed
S1-20	Order view returns 500 (MissingGreenlet)	M6	Defect	Lazy-loading of line items and products fixed
S1-21	Log the issues being faced	M8	Doc	Migration issue log created
S1-22	Proceed to the next step	M7	Dir	AI-native layer started
S1-23	Preference banner and favorites list appear unimplemented	M4	Defect	Category banner and Pet Favorites sidebar implemented
S1-24	Are the schema and APIs optimized compared with the source?	M8	Review	Async I/O, N+1 avoidance, indexing and REST tools reviewed
S1-25	Updating preferences must be honoured immediately	M5	Review	Profile changes verified to update banner, favorites and database
S1-26	Which phases are still pending?	M9	Q	Status check: phases 1 to 3 done, phase 4 pending
S1-27	Language preference (English, Japanese) does not work	M5	Defect	Dynamic localization implemented
S1-28	Harden order data APIs so details are not visible without user context	M6	Dir	Order access restricted to the owner; authenticated order APIs with card masking

ID	Prompt (reworded)	Tag	Type	Outcome
S1-29	Is localization implemented to standards?	M5	Review	gettext conventions, ISO 639-1 codes and Accept-Language negotiation
S1-30	New user registration fails (MissingGreenlet on profile after commit)	M5	Defect	Lazy-load after commit fixed
S1-31	What caches are implemented for search?	M7	Q	Search cache, catalog cache and precompiled concepts explained
S1-32	Pet Favorites display lacks parity with the original UI (screenshot attached)	M4	Defect	Side-by-side layout restored; missing product IDs fixed

4. Session 2 prompts

ID	Prompt (reworded)	Tag	Type	Outcome
S2-01	Proceed with Phase 4	M8	Dir	Verification and delivery phase started
S2-02	Write a detailed, concise document on deviations from the legacy system with rationales	M8	Doc	Deviations and rationales document
S2-03 to 05	Remove local filesystem paths, standardize links to repo-relative paths, consolidate docs under docs/	M8	Doc	Documentation links cleaned
S2-06 to 08	Fix redundant markdown link text across documents	M8	Doc	Duplicate link text removed
S2-09	Scrutinize the live demonstration guide	M8	Review	Demo guide audited

ID	Prompt (reworded)	Tag	Type	Outcome
S2-10 to 12	Resolve host and port mismatch between container (8000) and local run (8080); confirm readiness	M8	Defect	Ports and instructions made consistent
S2-13 to 14	Overhaul the README as the main evaluation deliverable, in an enterprise architect's tone	M8	Doc	README rewritten
S2-15	Verify there is no hallucinated or placeholder content in the documents	M8	Review	Documents checked
S2-16	Validate Docker build and run commands	M8	Review	Container build and run verified
S2-17	Verify the health check endpoint	M8	Review	GET /health confirmed
S2-18	Make Swagger UI, ReDoc and OpenAPI JSON links explicit in the README	M8	Doc	API documentation linked
S2-19	Review the authentication mechanisms for web and API	M5	Review	Current mechanisms assessed
S2-20	Point out ad-hoc session checks on protected endpoints (checkout, profile update)	M5	Defect	Risk of inconsistent access control identified
S2-21 to 22	Implement a central declarative auth dependency and session-fixation defense; record it as a deviation	M5	Dir	Central auth dependency added; Deviation 11 recorded
S2-23 to 24	Audit the remaining tasks against the assessment requirements	M9	Review	Outstanding work listed
S2-25	How can the Pet Advisor drawer be validated?	M7	Q	Validation steps provided

ID	Prompt (reworded)	Tag	Type	Outcome
S2-26	Remove hallucinated products and cryptic codes; ground recommendations in real catalog items	M7	Defect	Advisor grounded in database items
S2-27	Show real titles, descriptions, prices and a one-click Add to Cart in advisor cards	M7	Dir	Advisor card UX improved
S2-28	Personalize to the user's favorite category; add category pills	M7	Dir	Profile-based recommendations
S2-29	Replace the harsh error banner on casual queries with a polite fallback	M7	Defect	Conversational fallback added
S2-30	Draft an update to inform the hirer	M9	Doc	Update writeup
S2-31	Clarification: this is the formal submission report, not a feature update	M9	Dir	Writeup reframed as submission report
S2-32	Produce the submission report as a Word document	M9	Doc	Technical assessment submission report (.docx)
S2-33	Draft the handover email	M9	Doc	Handover email
S2-34	Summarize all prompts used	M9	Doc	Source of this log

5. Usage readings (Antigravity model pools)

Antigravity reports quota as a percentage per model pool, not as token counts. These readings are recorded as percentage points and are not converted to tokens or cost.

Reading	Gemini weekly remaining	Gemini five-hour remaining	Claude and GPT weekly remaining
28 Oct, 9:28 AM	89%	100%	0%

Reading	Gemini weekly remaining	Gemini five-hour remaining	Claude and GPT weekly remaining
28 Oct, 12:34 PM	80%	71%	0%
Change	9 points	29 points	none (already exhausted)

6. Observations from the log

- **Where the effort went.** 20 of 66 prompts (30%) were verification, documentation and deployment work, and 9 of 66 were strategy and ADRs. Decisions were made before code was generated.
- **Defect-driven iterations.** Prompts typed Defect are the iteration evidence: build errors (S1-14, S1-15), runtime errors (S1-17, S1-20, S1-30), and parity gaps (S1-19, S1-23, S1-27, S1-32). Parity gaps were found by visual comparison with the original.
- **Security hardening was prompted by review, not by the first draft.** S1-28 and S2-20 to S2-22 changed order access and session handling after review.
- **Grounding was added iteratively.** S2-26 removed hallucinated advisor items, which supports the guardrail that recommendations come from database items.

7. Limits of this evidence

- Model pool percentages are account-wide and cannot be attributed to a single model or milestone. The pool may include other work in the same account.
- The model used for each prompt was not recorded, and per-prompt token counts are not available from the IDE.
- Session 1 counts come from the session's own summary and were not re-extracted from the transcript. Session 2 was extracted from the transcript.
- Percentage readings were taken at two points only, so they are not split by milestone.